

Déjà Vu of Strange Stickers! Enhancing Out-of-Distribution Robustness in Sticker Retrieval via Cross-Modal Intent Alignment

Yu-An Liu

University of Chinese Academy of Sciences, CAS
Beijing, China
liuyuan21@mailsucas.ac.cn

Ruqing Zhang*

University of Chinese Academy of Sciences, CAS
Beijing, China
daqingchongzrq@gmail.com

Jiafeng Guo*

University of Chinese Academy of Sciences, CAS
Beijing, China
guojiafeng@mailsucas.ac.cn

Changjiang Zhou

University of Chinese Academy of Sciences, CAS
Beijing, China
zhouchangjiang23@mailsucas.ac.cn

Fan Zhang

WeChat Search Application
Department, Tencent Inc.
Guangzhou, China
fatumzhang@tencent.com

Yinhu Zhao

WeChat Search Application
Department, Tencent Inc.
Guangzhou, China
hugoyhzhao@tencent.com

Abstract

The rapid growth of digital communication has increased the demand for sticker retrieval systems that can match expressive stickers to users' communicative needs. In practice, however, sticker retrieval encounters significant out-of-distribution (OOD) challenges arising from unseen queries and stickers, driven by the diversity of user expression habits and sticker visual representations. These OOD issues often lead to irrelevant or inappropriate retrieval results, undermining the user experience. Drawing on symbolic interactionism in cognition, we propose XAlign-SR, a method that enhances OOD robustness by aligning abstract expressive intent between queries and stickers across modalities. To support this study, we construct OOD benchmarks from sticker datasets that simulate realistic query-sticker scenarios. Experiments demonstrate that our approach significantly outperforms state-of-the-art baselines.

CCS Concepts

• Information systems → Retrieval models and ranking.

Keywords

Sticker retrieval, Cross-modal matching, OOD robustness

ACM Reference Format:

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*Jiafeng Guo and Ruqing Zhang are the corresponding authors.



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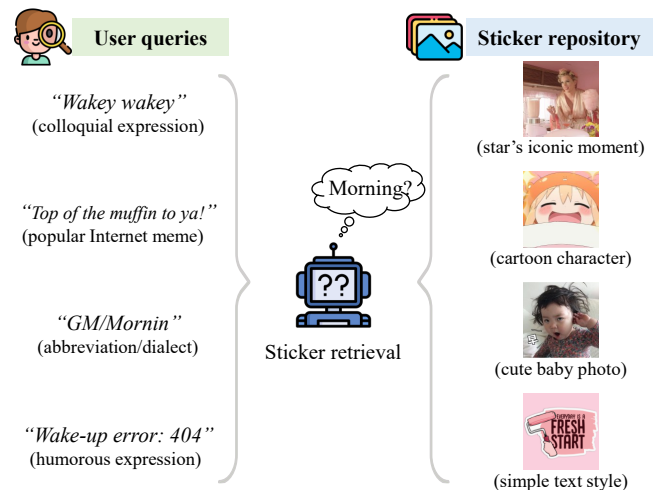


Figure 1: The sticker retrieval is challenged by the diverse query expressions and various sticker presentation formats, even under the same expressive intent.

1 Introduction

The popularization of instant messaging and social media platforms has cemented stickers as indispensable tools for digital communication. During online chat, stickers serve as vivid visual elements that enhance conversational dynamics by transcending linguistic and cultural barriers [39]. Effective sticker retrieval systems, which match user queries with stickers that align with their intended expression from a massive repository, are crucial for meeting users' evolving expressive needs in real-world applications.

Out-of-distribution challenge in sticker retrieval. Typical retrieval tasks such as web search [27] and patent retrieval [23] usually serve users' information needs. In contrast, sticker retrieval-integrated within instant messaging platforms-caters to users' expressive needs. Unlike the concreteness and accuracy of information needs, expressive needs are often abstract, diverse, and trend-driven. This unique scenario gives rise to distinctive characteristics of the query side and the sticker side, as shown in Figure 1: (i) *User queries are expressed in various ways.* Queries are usually short and casually expressed, leading to significant query diversity. This variation arises from personal expression, cultural background, and even

typing habits. During our investigation on online sticker search engines, user queries for “good morning” may appear in over 28 variations, including synonyms, abbreviations, typos, and incomplete phrases, “morrning,” “sunshine,” “gm,” “fresh day,” or even playful phrases like “wakey wakey.” (ii) *Stickers with the same expression intent can vary significantly*. This variation stems from differences in artistic design, cultural influences, and content. Take one of the most popular instant messaging platforms [9], WeChat [41], as an example, stickers expressing happiness involve over 169 characters and 18 styles, ranging from simple emoji, artistic text, to animated characters, and meme-inspired images. The ongoing evolution of internet culture also continuously introduces novel stickers that interest users, resulting in an ever-growing and diverse collection.

These characteristics collectively present out-of-distribution (OOD) challenges in sticker retrieval during training. OOD challenge refers to the distribution shift between the data encountered during testing and that seen during training, which leads to a performance degradation [14]. Specifically, it can be divided into two main scenarios: (i) *OOD queries* refer to unseen query expressions; and (ii) *OOD stickers* refer to unseen or newly popular sticker contents and styles. The inherent differences between query and sticker modalities, combined with the diverse expressions of queries and stickers, make the relevance pattern difficult to learn naturally, resulting in retrieving plenty of low-quality, irrelevant stickers in practical sticker retrieval. In this paper, we try to explore how to improve the OOD robustness of the sticker retrieval model and show how OOD robustness can help deploy an effective and reliable system and ultimately enhance user experience.

Using OOD retrieval solutions for sticker retrieval. Yet little effort has been directed toward addressing OOD challenges in sticker retrieval. The most related work in this direction has focused on OOD issues in text retrieval [12, 42, 51], where the key idea is to enhance a model’s fine-grained matching ability between queries and documents, allowing it to ignore spurious correlations and focus on truly relevant features. However, such approaches are not directly applicable to sticker retrieval, where relevance depends on capturing abstract expressive intent rather than fine-grained textual matching. As queries and stickers with the same intent can take diverse forms, strict fine-grained matching may introduce noise. Our experiments show that applying state-of-the-art OOD documents [15, 51] or OOD queries [56] methods from text retrieval yields only marginal gains in sticker retrieval (Section 5.4), as they struggle to capture the abstract intent between queries and stickers.

Our approach: Cross-modal intent alignment. In this paper, we propose a novel training method for sticker retrieval called XAlign-SR. Our goal is to mitigate the challenge of OOD queries and stickers by identifying the core expressive intent behind various query and sticker expressions, and aligning their cross-modal expressive intent during training. This approach is inspired by the cognitive science concept of symbolic interactionism [2], which posits that *humans abstract expressive intent by interactively aligning cross-modal expressive symbols (such as queries and stickers) in the brain*.

XAlign-SR trains a text intent encoder for queries and sticker text, and an image intent encoder for sticker images in three key steps: (i) Text-focused intent understanding, using chain-of-thought (CoT) reasoning to capture intent and generate diverse expressions

for contrastive learning; (ii) Image-focused intent understanding, jointly training the image and text encoders to align visual expressions of similar intents; (iii) Cross-modal intent alignment, aligning query and sticker representations across modalities to improve relevance. Expression-oriented distributional robust optimization (DRO) ensuring data efficiency and OOD robustness.

Experiments and contributions. Based on three open-source sticker retrieval datasets, i.e., WeChat [47], Sticker820K [54], and U-Sticker [6], we construct OOD robustness scenarios for experimental evaluation. Experiments demonstrate that our method outperforms state-of-the-art sticker retrieval models as well as OOD-robust retrieval baselines. To our best, we are the first to address the underexplored challenge of OOD sticker retrieval.

2 Related Work

Cross-modal retrieval. Cross-modal retrieval refers to retrieving relevant data across different modalities based on a query from one modality [45]. Text-to-image retrieval bridges the semantic gap between text queries and images [10, 34]. Existing approaches focus on object recognition [53], object relationship reasoning [30], and enhancing detail discernment via denoising or adversarial learning [22, 36, 38]. However, in sticker retrieval, sticker images often abstractly convey certain concepts or emotions [39]. Traditional image retrieval methods with detailed object recognition could be susceptible to the highly diverse ways stickers express meaning.

Sticker retrieval. Sticker retrieval has become a key component of instant messaging apps, enabling users to find stickers that match their expressive intent via text queries [54]. For instance, Zhao et al. [54] leveraged CLIP [31] to model sticker visual and textual features, while Int-RA [19] introduces relation-aware retrieval. Most prior work evaluates performance under IID settings; in contrast, we study a more realistic and challenging OOD setting, aiming to improve sticker retrieval robustness under distribution shifts.

OOD robustness of retrieval model. Neural retrieval models are known to be vulnerable to unseen data due to the limitations of deep neural networks [20, 37, 42]. Despite strong IID performance, they often degrade under OOD shifts [7, 29, 42]. Prior work improves OOD robustness mainly via data augmentation [3, 57], distributionally robust optimization [51], or domain-invariant projection [49]. These approaches generally encourage models to rely on fine-grained matching cues, mitigating spurious content variations. Sticker retrieval, however, poses a distinct OOD challenge: robustness depends more on aligning the abstract expressive intent of queries and stickers than on fine-grained lexical or visual matches, making existing methods less directly applicable.

3 Problem Statement

Task description. Given a textual query q , the aim of sticker retrieval is to return a ranked list $\mathcal{R}$ of top- K relevant stickers from a large sticker repository $\mathcal{S} = \{s_1, s_2, \dots, s_N\}$ with a total of N stickers, prioritizing the more relevant the closer to the top. In general, each sticker s consists of text t and image v , where the text includes the caption and style category, etc. The sticker retrieval model f should produce a relevance score $\text{Rel}(q, s)$ of the query q for each sticker s in $\mathcal{S}$, based on the sticker text t and image v .

Then a truncated ranked list $\mathcal{R}$ is recalled by selecting the top- K stickers with the highest relevance scores, where $K \ll N$.

OOD robustness in sticker retrieval. In retrieval tasks, the OOD robustness refers to retrieval models’ ability to generalize and maintain ranking performance when encountering unseen data [21, 42]. Formally, given a retrieval model $f_{\mathcal{D}_{\text{train}}}$ trained on the original training data $\mathcal{D}_{\text{train}}$, its OOD robustness is derived from its ranking performance R_M under unseen test data $\mathcal{D}_{\text{test}}^*$ with metric M :

$$\text{Robustness}_{\text{OOD}} = R_M(f_{\mathcal{D}_{\text{train}}}; \mathcal{D}_{\text{test}}^*). \quad (1)$$

In sticker retrieval, due to the diverse and abstract expression of queries and stickers, the unseen test data $\mathcal{D}_{\text{test}}^*$ is inherently divided into two types: (i) *OOD queries* refer to unseen query variations for models of the same expression intent, arising from typos, language habits, or memes et al; and (ii) *OOD stickers* refer to unseen expressions of the same intent, including strange or newly emerging content, style et al. In this paper, we propose a method to simultaneously address the challenges posed by OOD queries and stickers, while evaluating robustness in three OOD scenarios separately.

Sticker retrieval datasets. We collect three sticker retrieval datasets in English and Chinese: (i) **WeChat** [47], a public Chinese dataset from the CCIR Cup 2024 sticker retrieval challenge [5], containing 500K stickers spanning 73 styles and 6,250 intellectual properties (IPs; i.e., characters); (ii) **Sticker820K** [54], a public dataset with ~820K stickers across 8 styles; (iii) **U-Sticker** [6], a large-scale multi-domain dataset with 370K stickers across 10 domains, primarily in English. In WeChat, each sticker includes an image and metadata (caption, emotion, style, IP, and OCR text). We define the sticker text t as the concatenation of caption, emotion, style, IP, and OCR, and denote the sticker image as v . Sticker820K provides similar fields but without IP, while U-Sticker includes only captions.

OOD benchmark construction. Since these datasets do not have a direct OOD data part, we further construct OOD benchmarks for evaluation by dividing the query and sticker data. We construct the benchmarks of WeChat, Sticker820K, and U-Sticker datasets, respectively. The overall statistics are shown in Table 1.

For *OOD queries*, we randomly sample 5% of labeled queries from each of the WeChat and Sticker820K datasets to generate their variants as OOD queries. Specifically, we invited practitioners with experience working with sticker search to write five variants for each sampled query. Each sampled query was annotated by three practitioners, with another practitioner reviewing and analysing the annotations for potential bias. These variants simulate differences arising from typos, verbal expression habits, and cultural contexts, but are guaranteed to be relevant to the original stickers. For U-Sticker dataset, we leverage the dataset’s inherent multilingual nature by extracting the minority Chinese queries as test queries to evaluate cross-lingual OOD robustness. We exclude stickers associated with sampled queries from the training set and use them along with the query variants as the test set for the OOD queries.

For *OOD stickers*, we divide about 10% of the labeled stickers from each of the two datasets and ensure that types of style and IP of the divided stickers no longer exist in the original dataset. For Sticker820K, since the dataset does not contain an IP field in its metadata, we perform segmentation based only on style. For U-Sticker, we sample stickers from five niche domains—Arts, FanClub,

Outdoor, Social, and Tech—out of the dataset’s ten sticker domains, and use them as OOD stickers. The sampled stickers, along with their associated queries, are excluded from the training set and used as the test set for OOD evaluation.

Table 1: The data statistic of WeChat, Sticker820K, and U-Sticker. # Pairs denotes query-sticker pairs.

Scenario	Data	WeChat	Sticker820K	U-Sticker
Original	# Queries	11,018	50,235	230K
	# Stickers	450K	800K	230K
	# Pairs	11,519	50,302	230K
OOD Queries	# Queries	2,465	12,530	70K
	# Stickers	500K	800K	70K
OOD Stickers	# Queries	1,243	5,024	22k
	# Stickers	500K	800K	22K

Discussion. The OOD challenges in sticker retrieval differ from traditional text/image retrieval [4, 13, 46, 55]. Traditional retrieval focuses on matching key details to meet informational needs, with OOD issues often mitigated by keyword/object matching [42]. In contrast, sticker retrieval involves diverse visual factors and casual, expressive context, increasing OOD challenges. It prioritizes capturing query intent over literal content, as visual elements of stickers often can not directly serve as a reliable relevance criterion. To address this, methods must effectively capture the abstract intent behind both the user’s query and the stickers’ expressive nature.

4 Our Method

We introduce XAlign-SR for sticker retrieval OOD robustness.

4.1 Overview

Sticker metadata includes an image v and text t , which are encoded independently with separate encoders; the text encoder also encodes queries. As shown in Figure 2, XAlign-SR consists of three components: (i) **Text-focused intent encoder**, capturing expressive intent from queries and sticker text; (ii) **Image-focused intent encoder**, capturing the visual expressive intent of stickers; (iii) **Cross-modal intent alignment**, interactively aligning query and sticker intents to enable relevance matching. We further apply **expression-oriented distributionally robust optimization** throughout XAlign-SR training.

4.2 Text-Focused Intent Understanding

The goal of this stage is to address OOD issues arising from text modalities by accurately understanding the expressed intent of both the query and sticker text. We start by warming up the text-intent encoder (see Section 4.5) and then further refine it to enhance its intent identification capabilities. Specifically, we first expand the short query and then train the text encoder using a contrastive loss.

CoT-based query expansion. Queries in sticker retrieval are typically shorter and more casual than in standard retrieval tasks. To address this, we use LLMs to infer the underlying expressive intent and enrich each query with alternative paraphrased expressions.

Given a query q (which may be an explicit expression or a direct intent): (i) We prompt an LLM to analyze q . If q is an expression, the LLM extracts the underlying intent i_q (Appendix A.1.1); otherwise, we treat the query itself as intent and continue. (ii) Conditioned on i_q , we use a CoT-style generation process [44, 48] to produce

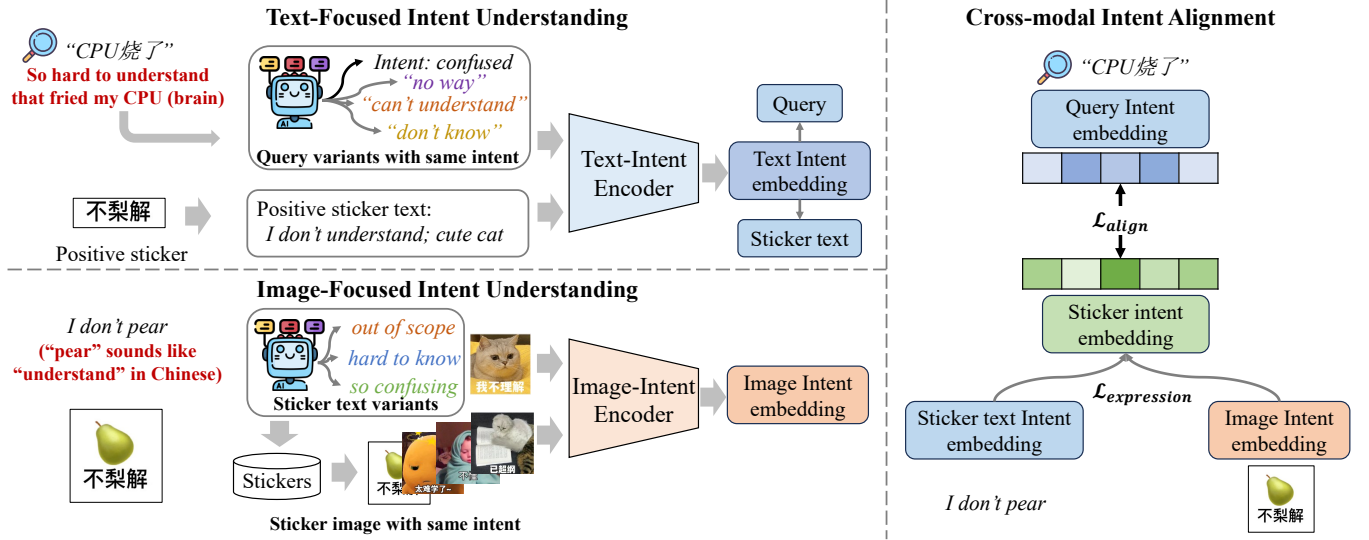


Figure 2: The framework of XAlign-SR that understands and aligns the expressive intent of queries and stickers.

k alternative expressions. Each step generates a new paraphrase based on the intent and previously generated ones (Appendix A.1.2), yielding $e_1, \dots, e_k$ with shared intent but varied wording. (iii) We then form an enriched query by concatenating the original query, intent, and generated expressions: $q' = [q; i_q; e_1; \dots; e_k]$.

Text-intent encoder. An effective text encoder should cluster different textual expressions, including queries and sticker text, that share the same intent in semantic space. For each expanded query and its corresponding positive sticker pair (q', s^+) , where sticker $s^+ = (t^+, v^+)$ contains sticker text and image: (i) We first use LLMs to generate n variant textual expressions $\{t_1^+, \dots, t_n^+\}$ for the original sticker text t^+ ; (ii) Then, we sample m random sticker texts $\{t_1^-, \dots, t_m^-\}$ as negative examples; and (iii) Finally, we train the text encoder to capture the expressive intent of queries and sticker text within the output text embeddings using contrastive loss:

$$\mathcal{L}_{\text{text}} = -\log \frac{\sum_{i \in n} \exp(\mathbf{q}^\top \mathbf{t}_i^+)}{\sum_{i \in n} \exp(\mathbf{q}^\top \mathbf{t}_i^+) + \sum_{j \in m} \exp(\mathbf{q}^\top \mathbf{t}_j^-)}, \quad (2)$$

where $\mathbf{q}$ and $\mathbf{t}$ are intent embeddings of query and sticker text, respectively, generated by text encoder.

4.3 Image-Focused Intent Understanding

The aim of this stage is to train an image-intent encoder to mitigate OOD issues arising from diverse visual expressions in sticker images. We begin by warming up the image-intent encoder (see 4.5) and then enhance its intent identification capabilities.

The image-intent encoder should cluster diverse visual representations that share the same intent in semantic space. Given an expanded query and its corresponding positive sticker pair (q', s^+) : (i) We match the query-sticker text pair in the training set using the n variant textual sticker expressions $\{t_1^+, \dots, t_n^+\}$ generated above; (ii) For each generated expression t_i^+ , we compute the cosine similarity using the text-intent encoder and take the image embedding of the most similar query-sticker text pair as the positive examples $\{\mathbf{v}_1^+, \dots, \mathbf{v}_n^+\}$. Additionally, we randomly sample m sticker images $\{\mathbf{v}_1^-, \dots, \mathbf{v}_m^-\}$ from the sticker repository as negative examples; and

(iii) Finally, we train image encoder to capture the expressive intent of sticker images by optimizing the output image-based intent embeddings with contrastive loss:

$$\mathcal{L}_{\text{img}} = -\log \frac{\sum_{i \in n} \exp(\mathbf{v}^\top \mathbf{v}_i^+)}{\sum_{i \in n} \exp(\mathbf{v}^\top \mathbf{v}_i^+) + \sum_{j \in m} \exp(\mathbf{v}^\top \mathbf{v}_j^-)}, \quad (3)$$

where $\mathbf{v}$ denotes the intent embedding of the sticker image, generated by the image encoder.

4.4 Cross-Model Intent Alignment

After learning the respective intents of text and images, the next challenge is to effectively match the intents expressed by the query and the sticker for meaningful relevance. Specifically, we propose cross-modal intent alignment, where query intent embeddings are aligned with sticker intent embeddings that convey the same intent during training. This approach enables the model to learn query and sticker intent embeddings that not only recognize intent but also effectively match it across modalities.

Given an expanded query and its corresponding positive sticker pair (q', s^+) , the cross-modal intent alignment is performed as follows: (i) *Extracting sticker embeddings:* We compute the sticker text embedding $\mathbf{t}_s$ and the sticker image-based intent embedding $\mathbf{v}_s$ using the text and image encoders:

$$\mathbf{t}_s = f_T(t), \quad \mathbf{v}_s = f_I(v), \quad (4)$$

where f_T and f_I represent the text and image encoders, respectively.

(ii) *Integrating sticker representations:* We employ a Transformer [43] where sticker image-based intent embeddings $\mathbf{v}_s$ serve as values (V) and keys (K), while sticker text intent embeddings $\mathbf{t}_s$ serve as queries (Q), producing an integrated sticker intent embedding $\mathbf{h}_s$:

$$\mathbf{h}_s = \text{softmax} \left(\frac{QK^\top}{\sqrt{d}} \right) V = \text{softmax} \left(\frac{\mathbf{t}_s \mathbf{v}_s^\top}{\sqrt{d}} \right) \mathbf{v}_s, \quad (5)$$

where d is the embedding dimension and $\sqrt{d}$ is scaling factor. (iii) *Sticker expression Loss*: To ensure consistency between the text-based intent embedding and the integrated sticker intent embedding, we minimize the sticker expression loss:

$$\mathcal{L}_{\text{expression}} = \|\mathbf{t}_s - \mathbf{h}_s\|. \quad (6)$$

(iv) *Cross-modal alignment learning*: To align the expression intent of the query and sticker while distinguishing them from unrelated stickers, we employ cross-modal intent alignment loss with randomly sampled negative stickers:

$$\mathcal{L}_{\text{align}} = -\log \frac{\exp(\text{sim}(\mathbf{q}, \mathbf{h}_s))}{\sum_{s \in \mathcal{B}} \exp(\text{sim}(\mathbf{q}, \mathbf{h}_s))}, \quad (7)$$

where $\mathcal{B}$ is the training batch, $\mathbf{h}_s^-$ is negative sticker embedding, and $\text{sim}(\cdot)$ is the cosine similarity.

4.5 Expression-Oriented DRO

As analysed previously, the OOD challenge in sticker retrieval primarily stems from the diverse expressions of queries and stickers. Inspired by distributional robust optimization (DRO), which shows that models trained with balanced domains exhibit greater robustness in new domains [33, 51], we propose expression-oriented distributional robust optimization to train XAlign-SR. The training consists of three phases: unsupervised warm-up, semi-supervised joint intent learning, and expression-oriented DRO.

Unsupervised warm up. For the text and image intent encoders, we begin with an unsupervised training method to initialize both encoders simultaneously, ensuring they possess initial text-image matching capabilities. For each sticker s , we randomly sample one type of sticker metadata (i.e., captions, OCR text, IP tags) t'_s as query, and calculate the initialize loss with its image v_s :

$$\mathcal{L}_{\text{init}} = -\log \frac{\exp(\text{sim}(\mathbf{t}'_s, \mathbf{v}_s))}{\sum_{s \in \mathcal{B}_w} \exp(\text{sim}(\mathbf{t}'_s, \mathbf{v}_s))}, \quad (8)$$

where $\mathcal{B}_w$ represents the warm-up batch and $\mathbf{t}'_s$ is the embedding of the sampled sticker metadata.

Semi-supervised joint intent learning. After initializing the encoders, we propose jointly optimizing the intent understanding capabilities of the text-intent encoder (Eq. 2) and the image-intent encoder (Eq. 3). We dynamically balance the optimization process between the two encoders using adaptive weighting:

$$\mathcal{L}_{\text{joint}} = \alpha \mathcal{L}_{\text{text}} + (1 - \alpha) \mathcal{L}_{\text{img}}, \quad (9)$$

where $\alpha = \frac{\mathcal{L}_{\text{text}}}{\mathcal{L}_{\text{text}} + \mathcal{L}_{\text{img}}}$ is the adaptive weight. During training, we generate pseudo-labels using LLM and text similarity matching, enabling semi-supervised learning for the training data.

Expression-oriented DRO. Once the text and image encoders have acquired intent understanding capabilities, we first cluster the text-image expression $[\mathbf{q}; \mathbf{t}_s; \mathbf{v}_s]$ of query-sticker training pairs with K-Means [1], and then optimize the eDRO loss:

$$\mathcal{L}_{\text{eDRO}}(\theta) = \sum_{k=1}^K \omega_k \ell_k(\theta), \quad (10)$$

$$\ell_k = \gamma \mathcal{L}_{\text{expression}} + \mathcal{L}_{\text{align}}, \quad (11)$$

where γ is a weight hyperparameter, K is total cluster numbers set by the domain numbers of different datasets, θ are parameters of XAlign-SR, and ω_k is the weight of k -st cluster.

During training, ω_k is dynamically updated to maintain the model's adaptation to the distribution of expressions. Following [51], at training step c , the optimal weight ω_k^{t*} for each cluster can be approximated by Lagrangian multiplier and calculated as:

$$\omega_k^{t*} = \frac{\omega_k^{t-1} \exp(\sum_{x=1}^K r_{xy})}{\sum_{y=1}^K \omega_k^{t-1} \exp(K_{x=1} r_{xy})}, \quad (12)$$

$$r_{xy} = [\ell_x(\theta) \ell_y(\theta)]^\beta (\nabla_\theta \ell_y(\theta))^\top \ell_x(\theta), \quad (13)$$

where $\beta = 0.25$ is a hyperparameter that emphasize on harder clusters, and r_{xy} is the loss similarity between different clusters. It encourages model to capture the “common needs” shared across expression as the underlying intent, thereby enhancing robustness.

5 Experiments

We conduct experiments to verify the effectiveness of our method.

5.1 Evaluation

Evaluation scenario. For each benchmark dataset, we evaluate the retrieval performance across three scenarios, i.e., the *original scenario*, *OOD queries*, and *OOD stickers*, which correspond to the original retrieval effectiveness and the OOD robustness when faced with unseen queries and stickers. The model is trained under the training set that is IID with the original scenario.

Evaluation metrics. In retrieval tasks, OOD robustness is measured by the retrieval performance under OOD data [20, 42]. In this paper, we adopt two metrics for each scenario: (i) Mean reciprocal rank (MRR@K) measures how high the relevant stickers rank within the top-K result [24, 26]; and (ii) Recall@K measures the proportion of relevant stickers that exist in the top-K result [13, 25]. For ease of observation, all metrics are presented as percentages.

5.2 Baseline methods

Baselines. We compare our method with several representative approaches, including regular sticker retrieval models, retrieval methods tackling OOD queries and OOD documents (stickers, in this paper), respectively, from text & image retrieval:

- For regular sticker retrieval models, we adopt: (i) BM25 [35] is a classical probabilistic retrieval model that shows effectiveness on OOD retrieval tasks [42]. We take sticker text as the document; (ii) StickerCILP [54] directly fine-tunes CLIP model [31, 50] to capture sticker features and text features; (iii) StickerLLM [54] is similar with StickerCILP but uses ChatGLM-6B [52] as the query encoder; and (iv) Int-RA [19] matches stickers by understanding the common-sense requirements of the query.
- For retrieval methods that enhance robustness to OOD queries, we adopt: (i) DRTA [56] is from text retrieval and uses contrastive learning to bridge the gap between the query and the possible variants; (ii) DST [40] is from text retrieval and aligns the ranking list between query and variants in a self-training manner; and (iii) PlugIR [16] is from image retrieval and leverages LLM reasoning to understand and refine unseen queries.
- For retrieval methods that enhance robustness to OOD stickers, we adopt: (i) Inpars [15] is from text retrieval and generates

Table 2: Retrieval performance of XAlign-SR and the baselines across the original, OOD queries, and OOD stickers scenarios on two benchmark datasets; * indicates significant improvements over the best baseline ($p \leq 0.05$).

Dataset & method	Original Scenario			OOD Queries			OOD Stickers		
	MRR		Recall	MRR		Recall	MRR		Recall
	@10	@5	@10	@10	@5	@10	@10	@5	@10
WeChat									
BM25	19.1	20.5	23.3	8.3	9.6	12.3	16.7	17.9	20.9
StickerCILP	23.5	25.8	29.3	11.2	13.0	16.5	8.2	9.1	11.2
StickerLLM	25.7	27.3	31.2	16.5	18.6	21.3	12.5	14.1	17.9
Int-RA	32.8	34.8	39.9	19.3	20.7	24.6	16.9	18.0	23.3
DRTA	26.3	28.8	32.2	22.6	24.3	31.2	16.4	17.6	22.3
DST	26.5	28.3	32.6	23.2	24.9	30.1	16.2	17.8	22.8
PlugIR	28.6	31.9	36.8	25.2	27.0	32.9	18.7	20.6	24.3
DAR	24.8	26.2	30.6	18.2	19.9	23.4	19.9	21.8	25.6
COCO-DR	25.0	26.9	31.5	20.3	22.1	25.6	23.1	23.5	26.9
InPars	27.5	29.3	32.0	21.7	22.9	26.8	24.6	24.9	28.7
XAlign-SR (Ours)	35.6*	39.2*	43.1*	28.8*	32.3*	36.2*	27.0*	31.2*	35.4*
Sticker820K	MRR	Recall		MRR	Recall		MRR	Recall	
	@10	@5	@10	@10	@5	@10	@10	@5	@10
BM25	38.5	46.7	52.6	20.5	24.8	30.2	33.8	41.7	46.9
StickerCILP	52.1	67.1	72.8	29.2	41.6	49.8	26.3	30.2	39.8
StickerLLM	59.8	73.6	78.4	35.8	44.9	52.1	32.8	36.9	42.3
Int-RA	65.9	82.3	86.3	39.5	47.3	59.0	37.0	42.1	47.7
DRTA	60.8	75.2	81.3	49.8	64.0	67.3	38.4	53.0	60.2
DST	61.3	76.0	82.9	50.3	64.2	68.9	37.8	52.4	59.9
PlugIR	63.2	79.3	85.2	54.8	69.9	74.3	42.9	56.7	64.3
DAR	58.9	71.1	76.9	38.6	47.3	52.3	50.3	59.3	67.2
COCO-DR	60.3	74.3	79.6	41.2	52.8	57.9	53.1	66.3	70.2
InPars	63.0	78.6	84.6	43.8	56.7	62.1	55.6	69.8	74.3
XAlign-SR (Ours)	70.1*	89.6*	92.1*	60.3*	77.9*	82.3*	58.7*	74.8*	80.1*
StickerQueries	MRR	Recall		MRR	Recall		MRR	Recall	
	@10	@5	@10	@10	@5	@10	@10	@5	@10
BM25	31.5	40.1	66.8	19.3	24.7	42.9	25.8	33.5	55.7
StickerCILP	42.1	58.2	92.3	27.8	43.0	67.4	20.3	26.1	49.2
StickerLLM	48.3	64.8	98.1	33.9	47.1	70.8	24.7	31.9	52.3
Int-RA	52.7	72.5	100.0	36.5	48.3	79.6	28.9	37.2	59.1
DRTA	49.8	67.0	100.0	47.1	66.4	92.3	29.3	47.8	74.1
DST	50.3	66.9	100.0	48.4	65.8	93.2	28.6	46.3	74.8
PlugIR	51.7	69.4	100.0	52.9	72.5	100.0	33.1	49.7	79.9
DAR	47.3	61.8	96.7	37.6	49.1	70.2	38.5	51.8	83.4
COCO-DR	48.9	65.3	99.8	39.7	54.6	78.3	40.2	58.4	86.8
InPars	50.2	68.7	100.0	41.9	58.9	84.2	42.3	61.7	91.9
XAlign-SR (Ours)	56.8*	78.3*	100.0	57.6*	80.4*	100.0	44.3*	65.8*	98.6*

pseudo query for OOD stickers with LLMs; (ii) COCO-DR [51] is from text retrieval and uses distributionally robust optimization to learn domain features of unseen stickers; and (iii) DAR [22] is from image retrieval and uses a diffusion model to generate unseen stickers to assist with retrieval.

Variants of XAlign-SR. We also implement three variants of XAlign-SR for ablation studies: (i) XAlign-SR_{Text}, which removes the text-focused intent understanding and only uses the original query-sticker pair to train the text-intent encoder in Section 4.2;

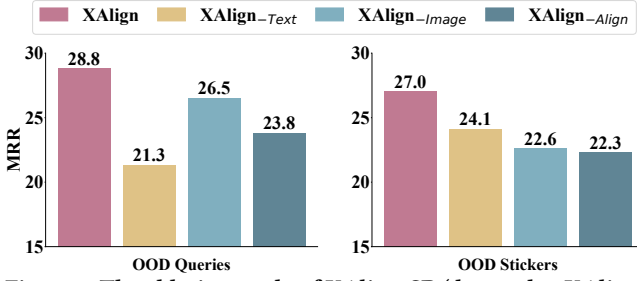


Figure 3: The ablation study of XAlign-SR (denoted as XAlign) in OOD queries and OOD stickers scenarios.

(ii) XAlign-SR-Image, which removes the image-focused intent understanding and only uses the original query-sticker pair to train the image-intent encoder in Section 4.3; and (iii) XAlign-SR-Align, which removes the cross-modal intent alignment step.

5.3 Implementation details

For the backbone model, following [26, 54], to facilitate comparison and account for online inference latency, both XAlign-SR and the baseline methods use CLIP [31, 50] as the text & image encoder. For the original scenario, we randomly extract 80% of the annotated query-sticker pairs for training and reserve the remaining 20% for testing. For sticker text, we directly contact the text fields of the sticker with the format of “Caption:, Emotion:, Style:, IP:, OCR:”. For the LLM applied in this paper, employ the *gpt-4-turbo* [28]. For XAlign-SR, we set the query variant number $k = 8$, the sticker text variant number $n = 7$, and the negative example number $m = 56$. During training, the loss hyperparameters α and γ are set to 0.6 and 0.01, respectively. We train the model with a batch size of 64, maximum sequence length of 256, and learning rate of $1e-5$.

5.4 Main result

Table 2 shows the comparison of XAlign-SR and baselines across original (IID) scenario and OOD scenarios.

Comparison in original scenario. From the in-distribution results, we observe that (i) WeChat performance is generally lower than Sticker820K because WeChat queries come from real users and are more diverse and noisy, highlighting the difficulty of real-world sticker retrieval. (ii) U-Sticker, built from chat logs, tends to have noisier annotations than WeChat, resulting in consistently lower performance. (iii) Sticker-specific methods (e.g., StickerCLIP and Int-RA) remain strong in this setting, and OOD-oriented training brings modest additional gains, suggesting that augmentation and robustness optimization can partly mitigate the noise and uncertainty inherent to sticker retrieval. (iv) XAlign-SR performs best overall by modeling the expressive intent of queries and retrieving stickers that convey the same intent.

Comparison in OOD scenarios. From the performance of baselines tailored for enhancing OOD robustness, we can observe that (i) BM25, which has demonstrated strong robustness in OOD text retrieval [29, 42], loses this advantage in OOD sticker retrieval, indicating that precise text matching cannot capture the abstract, intent-based relevance of stickers; (ii) StickerCILP/LLM, and Int-RA show significant performance drops in OOD settings, as they are tailored for IID scenarios and struggle to interpret the expressive intent behind novel queries and stickers; (iii) Methods targeting

Table 3: The MRR performance of XAlign-SR and representative baselines with different backbones under the scenario of original, OOD query, and OOD stickers, respectively.

Backbone	Method	Original	Query	Stickers
CLIP	PlugIR	28.6	25.2	18.7
	InPars	27.5	21.7	24.6
	Int-RA	32.8	19.3	16.9
	XAlign-SR	35.6	28.8	27.0
BLIP	PlugIR	32.3	28.5	21.5
	InPars	31.2	24.7	27.8
	Int-RA	37.1	22.0	19.6
	XAlign-SR	40.0	32.6	30.2
BLIP-2	PlugIR	33.9	29.6	22.9
	InPars	32.2	26.1	29.2
	Int-RA	38.3	23.3	20.9
	XAlign-SR	41.8	33.7	31.1

OOD queries or stickers improve robustness only in their respective scenarios, but underperform in the other. This suggests that focusing on a single modality is insufficient for comprehensive OOD adaptation in sticker retrieval; and (iv) Leveraging LLMs for OOD (e.g., PlugIR, InPars) outperforms similar approaches, as LLMs’ reasoning abilities better capture the expressive intent.

XAlign-SR outperforms strong baselines, showing that (i) Existing OOD techniques can be undermined by expressive noise in sticker retrieval because they over-emphasize fine-grained matching; XAlign-SR instead abstracts intent from both text and images, enabling intent-level relevance matching. (ii) Classical retrieval models often fail under OOD shifts due to the expressive gap between queries and stickers; XAlign-SR narrows this gap via cross-modal intent alignment. (iii) Jointly optimizing OOD robustness for both queries and stickers strengthens cross-modal learning, making XAlign-SR robust across modalities.

5.5 Ablation study

We compare XAlign-SR with three variants: XAlign-SR-Text, XAlign-SR-Image, and XAlign-SR-Align to validate the effectiveness of different components. The MRR performance in the original scenario of XAlign-SR, XAlign-SR-Text, XAlign-SR-Image, and XAlign-SR-Align is 35.6, 29.8, 32.9, and 27.5, respectively. The performance in OOD queries and stickers of XAlign-SR with its variants is shown in Figure 3. We report the MRR under WeChat dataset, with similar observations on the other dataset and metrics. We find that: (i) After removing intent understanding for the text and image modalities, the OOD performance for the corresponding query and sticker modalities significantly drops. The decline is more pronounced for the text modality, as it contains more information; and (ii) The cross-modal intent alignment component can further enhance OOD performance by building upon text/image intent understanding, achieving joint gains for OOD queries and stickers.

5.6 Backbone model analysis

Considering deployment costs in real-world scenarios, we follow existing work [19, 54] by adopting the lightweight CLIP model [32] as our backbone. To further validate our method’s performance across different backbones, we compare XAlign-XR and representative baselines on more powerful cross-modal pre-trained backbones.

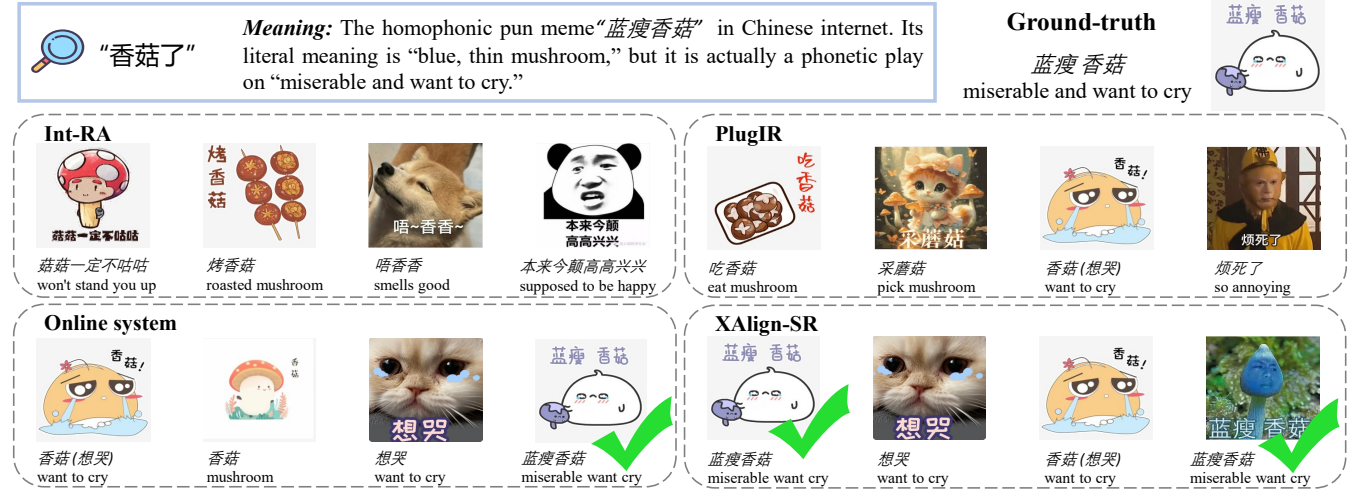


Figure 4: A case study on homophonic pun memes expressing sadness and the desire to cry across baselines and XAlign-SR.

Table 4: The user study between XAlign-SR, the online system, and representative baselines.

Method	MRR	Relevance	Preference
Online system	24.3	82.7	28.1
Int-RA	17.3	65.9	17.3
PlugIR	19.9	74.3	21.9
XAlign-SR	26.6	88.3	32.7

Specifically, we adapt our method to advanced backbones such as BLIP [18] and BLIP-2 [17]. We report the MRR under the WeChat dataset, with similar observations on the other dataset and metrics. As shown in Table 3, the BLIP series models consistently improve overall performance due to their strong backbones and pre-training capabilities. Notably, BLIP-2 demonstrates the best text and image understanding abilities, leveraging a powerful pre-trained ViT [11] and Flan-T5 [8]. Even as the backbone model changes, XAlign-XR maintains its performance advantage.

5.7 User study

To further validate the performance of XAlign-SR in real-world scenarios, we conducted a user study of our method in the sticker search system of WeChat. The dynamic sticker repository and the inherent unpredictability of user queries make sticker search a naturally OOD environment compared to the training set. We recruited 10 volunteers to conduct diverse daily queries, resulting in 151 valid searches. User clicks served as ground truth, and we selected a subset of the repository for evaluation. Users rated the overall relevance of results (0–100) and distributed 100 preference points among different model outputs. As shown in Table 4, XAlign-SR outperforms representative baselines and even online systems in both retrieval accuracy and user satisfaction, thanks to its robustness to OOD data and better alignment with user intent.

5.8 Case study

Figure 4 illustrates an example where a user searches for a popular internet homophonic meme sticker. In this case, the literal meaning of the query is “blue, thin mushroom”, but it actually represents a widely recognized meme expressing sadness and the urge to

cry. From the results, we observe that baseline methods typically rely on the literal meaning, retrieving multiple stickers related to mushrooms while also being misled by seemingly relevant but ultimately unrelated stickers. Online systems can partially capture the intended meaning of “feeling sad and wanting to cry”, but they fail to accurately retrieve stickers associated with the specific meme. In contrast, XAlign-SR effectively understands the user’s intent and the expressive meaning embedded within the meme stickers, enabling precise retrieval.

6 Conclusion

We propose XAlign-SR, a cognitive-inspired framework that aligns cross-modal expressive intent to perform robust relevance judgment under unseen data in sticker retrieval. We contrast two benchmarks for OOD sticker retrieval, including OOD queries and OOD stickers, to enable systematic evaluation of OOD robustness. Experiments demonstrate XAlign-SR outperforms baselines and online systems in terms of OOD scenarios. In future work, we will explore cost-effective model updates tailored to the evolving sticker scene. In practical scenarios, it is essential to reduce retrieval latency while fully leveraging a shared retrieval model to adaptively handle diverse and unknown data. Furthermore, we will explore adversarial robustness challenges, aiming to defend against malicious attacks that degrade system performance.

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A Appendix

A.1 LLM prompts

A.1.1 Prompt for CoT-based query expansion. The guiding prompt for recognizing the expressed intent is: “Given a user query for searching stickers, {query}, it may either represent an expressive form of intent or a direct intent expression. For example, “overjoyed to the point of flying” is an expressive form, whereas “happy” is a direct intent expression. First, determine whether the query is an expressive form of intent or a direct intent expression. (i) If it is an expressive form of intent, analyze its

underlying core intent and output it directly; (ii) If it is a direct intent expression, output it as is.”

The prompt for guiding the step-by-step generation of the different expressions of the query is: “Given a user query for searching stickers, along with its expressive intent, your task is to generate a new variant of the sticker query that aligns with the given intent while ensuring it is distinct from both the original query and any previously generated variants. The new variant should maintain the intended meaning but introduce differentiation wherever possible.

Query: {query}

Intent: {intent}

Pervious variants: {variants}”

A.1.2 Prompt for generating variant sticker text expressions. The prompt for generating variant sticker text expressions is: “Given a user query for searching stickers and the associated sticker text (including captions, text within the sticker, characters, and sticker style), the task is to identify the abstract expressive intent behind the query and the sticker text. Based on this understanding, a new variation of the sticker text should be directly generated that aligns with the given expressive intent while ensuring it is distinct from both the original text and previously generated variations. The generated variation should maintain meaningful differentiation while staying true to the intended expression.

Query: {query}

Relevant sticker text: {sticker text}

Pervious variants: {variants}”